

Phung Trong Hung

AI Engineer | LLM & Agentic Systems

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SUMMARY

AI Engineer who ships production LLM systems end to end. At UEF, built a RAG + agent learning assistant for Canvas LMS (92% answer accuracy, 1,000+ concurrent users) and, as sole AI engineer, a staff-portal assistant with 43 permission-scoped tools. First author of two papers: *Physical Review A* (submitted) and ICT 2026 (oral presentation).

SKILLS

Languages:	Python, C#, TypeScript, JavaScript
LLM & Agents:	LangGraph, LangChain, Semantic Kernel, Tool Calling, MCP, Human-in-the-loop workflows, OpenAI / Gemini / vLLM / Ollama, Speech-to-Text
Retrieval (RAG):	RAG, Graph-RAG, Hybrid search (BM25 + vector) + RRF, Query rewriting, Reranking, Docling, ChromaDB, pgvector, Evaluation sets
Backend & DevOps:	FastAPI, .NET (ABP), Next.js, React, Electron, PostgreSQL, Docker, GitHub Actions, Blue/green CI/CD
ML & Research:	PyTorch, TensorFlow, Transformers, Hugging Face, OpenCV, YOLO, Qiskit, SCIP

WORK EXPERIENCE

Department of Information Technology and Digital Transformation - University of Economics and Finance (UEF) Apr 2026 - Present

AI Engineer

UEF Learning Assistant - AI chatbot for Canvas LMS (LangGraph, FastAPI, ChromaDB, PostgreSQL)

- Built and deployed a LangGraph RAG + agent chatbot embedded in Canvas LMS (LTI 1.3) and Zalo OA; improved answer accuracy from 78% to 92% at p95 latency <2s, supporting 1,000+ concurrent users.
- Engineered the retrieval pipeline: Docling parsing, contextual chunking, query rewriting/expansion, hybrid BM25 + vector search fused with RRF, context compression, and a web-search fallback restricted to official UEF sources.
- Added agent tools over Canvas APIs that answer grades, deadlines, and submission status across all enrolled courses in one step; routed across Gemini, OpenAI, FPT vLLM, and Ollama; shipped via blue/green CI/CD with automated backups.

UEF Workplace - AI Assistant for the internal staff portal (sole AI engineer; .NET 10/ABP, Next.js, pgvector, OpenAI)

- Built an assistant that answers questions on internal regulations with mandatory citations and runs 43 permission-scoped tools (calendar, rooms, vehicles, leave, timekeeping, request approval), with human confirmation required for every write.
- Designed proactive background agents (schedule conflicts, overdue requests, expiring documents) that feed a Task Center, backed by fail-closed autonomy, outbox/inbox events, and compensating workflows.
- Masked PII and blocked HR/payroll data before any cloud LLM call; covered the module with 760+ backend and 1,600+ frontend tests.

UEF Meeting Notes - AI meeting minutes & live translation (sole developer; Next.js, Electron, OpenAI & Gemini speech APIs)

- Shipped a web and desktop app (Windows/macOS) that turns meetings of up to 3 hours into speaker-labelled transcripts and editable minutes exported to Word, with cited RAG Q&A over each meeting.
- Built real-time bilingual speech translation with floating subtitles over any app (Zoom, Teams, PowerPoint), an AI post-editing pass, and per-model API cost tracking.

UEF IT Management Center - University of Economics and Finance (UEF) Dec 2025 - Mar 2026

AI Engineer Intern

- Built RAG and Graph-RAG systems with Semantic Kernel (.NET) for university knowledge retrieval, including knowledge-base structuring and vector database design.
- Tuned retrieval to improve chatbot accuracy and latency for students and staff.

PUBLICATIONS & RESEARCH

Representation Dependence of Optimal Cut Placement in Quasiprobability Circuit Cutting Submitted Oct 2026

T. H. Phung, H. Bui. Submitted to *Physical Review A*, 2026. Code & data: Zenodo DOI 10.5281/zenodo.23205464

- Showed that equivalent gate-level representations of the same quantum circuit can change the resource-optimal cut placement under a QPD cost model; in one $n=14$ QPE case, reusing the other representation's optimum raises sampling overhead $729\times$ (confirmed by exhaustive audit of all 1,716 bipartitions).
- Formulated placement as weighted graph partitioning solved exactly with SCIP; released a reproducible pipeline (Qiskit, MQT Bench/QCEC equivalence checks, pinned environment).

PalmVeinFormer-TTA: ROI-free Palm Vein Recognition from Raw NIR Images with Transformer and Test-Time Augmented Gallery Scoring	May 2026
T. H. Phung , T. D. Nguyen, C. T. Tran, V. Q. Hoang, T. T. Nguyen, D. H. Bui. National Scientific Conference ICT 2026 - <i>Information and Communication Technology in the Era of AI and Comprehensive Digital Transformation</i> , Tien Giang University. Oral presentation.	
- Proposed a ROI-free recognizer that works directly on raw NIR images: CLAHE preprocessing, a Swin-Tiny encoder with ArcFace-style metric learning, and identity-level gallery scoring with probe-side test-time augmentation (TTA) and z-normalization. - Achieved EER 0.25%, Rank-1 99.26%, and TAR 98.89% at FAR = 1e-4 on a 1,549-identity in-house dataset (15,490 images); generalized zero-shot to the public Tongji raw dataset at EER 0.67%. - Runs at 11 ms/image (~88 images/s) in single-pass mode, with TTA as an optional high-security mode at ~30 ms/image.	

Earlier research: ECG heart-disease detection, Explainable AI, Event-camera vision	Mar 2025 - Dec 2025
Deep learning classifiers for ECG images with XAI (Grad-CAM, SHAP, LIME, saliency maps) for interpretability and error analysis; preprocessing and benchmarking of event-camera (DVS) data against frame-based methods.	

EDUCATION

Ho Chi Minh City University of Technology (HUTECH)	Oct 2022 - Jul 2026
Information Technology – Applied Machine Learning	

CERTIFICATIONS

Java Essentials 1 - Cisco Networking Academy	Dec 2025
JavaScript Essentials 2 - Cisco Networking Academy	Dec 2025
Networking Basics - Cisco Networking Academy	Nov 2025
Certificate in Communication and Teamwork Skills	Jun 2025
English Level B1 Certificate	Dec 2024